IBEH ESTHER IFEOMA

Full-Stack Software Developer

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PROFESSIONAL SUMMARY

Full-Stack Software Developer with a First Class Honours degree in Biochemistry, building maintainable, well-tested web applications across the stack with React, Next.js, TypeScript, and JavaScript on the frontend and Python/Flask, RESTful APIs, and SQL databases on the backend. Applies core engineering principles — separation of concerns, modular and extensible architecture, reusable components, and automated testing — to deliver reliable software. Built a 9-step fintech loan application with asynchronous verification and a Playwright end-to-end test suite, a widget-based portfolio analytics dashboard with a plugin-style component registry and financial calculations (Value-at-Risk, Pearson correlation, Brinson attribution) implemented from first principles, and a RESTful Task Management API in Python/Flask with an ORM-backed SQL database, input validation, automated Pytest coverage, and a live production deployment. Currently a remote contract developer on AmpFlyer, a multi-tenant event registration platform, building authentication and session flows and role-aware event management screens against a live REST API in a collaborative team monorepo. Experienced in the full development workflow: component and API architecture, database design, state management, version control, code quality, and continuous deployment. Detail-oriented and analytical, with prior experience in a clinical laboratory setting.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript (ES6+), Python, HTML5, CSS3
- **Frameworks & Libraries:** React, Next.js (App Router), Flask, Node.js, Express.js, Vite, Tailwind CSS, CSS Modules, Recharts, dnd-kit
- **Backend & Databases:** Flask, Flask-SQLAlchemy (ORM), RESTful API Design, SQLite, SQL, Pytest, Gunicorn
- **Engineering Practices:** Component-Based Architecture, Design Patterns, Separation of Concerns, Modular Design, State Management (Context API), Unit & End-to-End Testing, Debugging, Code Quality, Git Branching Workflow, Conventional Commits, CI/CD, Pull Request Workflow, Team Collaboration (GitHub Issues), Monorepo Development
- **Web Fundamentals:** RESTful API Design & Integration, Asynchronous Programming (Promises, async/await), Form Validation, Responsive Web Design, Web Accessibility (Semantic HTML, ARIA), Performance Optimization, Client-Side Storage (localStorage), Authentication & Session Management (JWT access/refresh tokens), Role-Based UI Access Control, API Error Handling
- **Tools:** Git, GitHub, Playwright, Storybook, Vercel, Render, npm, VS Code, Chrome DevTools, WSL, Canvas API, GitHub Issues, Monorepo (apps/packages)
- **Domain:** Statistical & Financial Modeling (VaR, Correlation, Performance Attribution)

PROJECTS

Multi-Step Fintech Loan Application | *React, Next.js, CSS Modules, Playwright*

- Architected a 9-step guided application form with centralized state management, per-step validation, and conditional field rendering across three loan types.
- Implemented asynchronous PAN verification and debounced address autocomplete using async/await, handling loading and response states to simulate real API interactions.
- Built document upload with type-aware previews and client-side image compression (Canvas API), reducing image size by roughly 90% without a backend, and a hand-drawn e-signature capture feature using React refs (useRef) and the same Canvas API.
- Engineered auto-save and resume with localStorage, diagnosing and fixing a race condition between save and load effects.
- Developed a rules-based loan eligibility engine and a live pre-approval summary driven by application state.
- Designed a custom design system (CSS Modules, design tokens, responsive layout) applied consistently across all steps, and authored an end-to-end Playwright test suite covering the full user journey, validation, and business-logic edge cases, resolving a hydration-timing race condition affecting test reliability.

Portfolio Analytics Dashboard | *React, Next.js, Tailwind CSS, Recharts, Storybook*

- Architected an extensible widget-based dashboard using a type-to-component registry (plugin-style architecture), allowing new widgets to be added without modifying core rendering logic.
- Implemented five analytics modules — Net Asset Value, Value-at-Risk, drawdown, Pearson correlation, and Brinson performance attribution — building the statistical calculations (covariance, standard deviation, running-peak tracking) from first principles in JavaScript.

- Built a mock data simulation layer using a random-walk algorithm to generate realistic time-series data, decoupled from the UI, and visualized results with Recharts.
- Implemented drag-and-drop widget reordering (dnd-kit) with global Context-based state managing a user-configurable layout.
- Built a Storybook component library with isolated stories for every major component, plus a Context-based theming engine for white-label configuration.
- Achieved 100/100 Lighthouse Performance and 98/100 Accessibility scores; established a Git branching workflow with Conventional Commits and continuous deployment via Vercel.

Task Management API | *Python, Flask, Flask-SQLAlchemy, SQLite, Pytest, Gunicorn, Render*

- Designed and built a RESTful task-management API in Python/Flask implementing full CRUD operations (GET, POST, PATCH, DELETE) with correct HTTP status codes and structured JSON responses.
- Modeled and persisted data with Flask-SQLAlchemy (ORM) over a SQLite database, replacing in-memory storage with a relational schema and auto-incrementing primary keys.
- Implemented server-side request validation and error handling (400/404 responses) to guard against missing, empty, or malformed input.
- Refactored into an app-factory architecture with Flask Blueprints and separated models, routes, and extensions modules, eliminating circular imports and enabling isolated test configuration.
- Wrote an automated Pytest suite using an in-memory SQLite database and Flask's test client to verify CRUD behavior and error handling independent of production data.
- Deployed the API to production on Render using Gunicorn as the WSGI server, with environment-based configuration for dynamic port binding.

Craps Game | *JavaScript, Python*

- Developed an interactive browser-based craps game implementing dice-roll logic, win/loss rules, and game-state handling in JavaScript, after first prototyping and validating the core game logic and betting rules in Python during the Simvo Full Stack Masterclass.

Additional Projects | *JavaScript, REST APIs*

- **Weather App, Quote Generator, BMI Calculator:** Responsive, API-integrated tools covering live weather forecasts, dynamic quote fetching, and BMI calculation.

WORK EXPERIENCE

Software Developer (Contract, Remote) | AmpFlyer Event Platform — Jul 2026 – Present

- Web developer on AmpFlyer, a multi-tenant event registration platform, working in a shared Next.js/TypeScript monorepo from a PRD and issue-based specs, with a pull-request workflow and CI checks.
- Built public sign-up, email verification (code entry and emailed link), and sign-in flows integrated with a live REST authentication API, with inline field validation, resend cooldown, redirects that steer unverified users to verification, and graceful handling of 400, 401, 403, 409, 410, and 429 responses.
- Implemented a token-based session layer: in-memory access token, persisted refresh token, automatic refresh-and-retry-once on 401, and logout that clears local state, so failures never leave a half-authenticated session.
- Built organizer event management screens (list, create, edit, detail) on a documented events API: paginated list with status filtering and archived toggle, validated create/edit forms with API error-code-to-field mapping, confirm-before-archive, and loading, empty, and error states.
- Applied role-aware UI (write controls limited to Owner, Admin, and Event Manager roles) and organization-scoped routing aligned with the platform's tenant-isolation model, surfacing 403 and 404 responses cleanly.
- Extended the shared typed API client, reused shared UI components, and added tests for form validation and error-code mapping.

Phlebotomist (NYSC) | Reddington Hospital — Jul 2025 – Jul 2026

- Performed blood draws and specimen collection with strict accuracy and adherence to safety and hygiene protocols.
- Maintained precise records under time pressure and communicated clearly with patients and clinical staff, building data-accuracy and process-discipline skills applied in structured development workflows.

EDUCATION

B.Sc. Biochemistry — First Class Honours (GPA: 4.59) | Covenant University, Ota, Ogun State — Sep 2020 – Oct 2024

CERTIFICATIONS

- Simvo Full Stack Masterclass (including Python-based application logic and problem-solving modules); Mimo Course Completion Certificates: TypeScript, JavaScript, React, HTML, Python